Hong Nguyen

Curriculum Vitae

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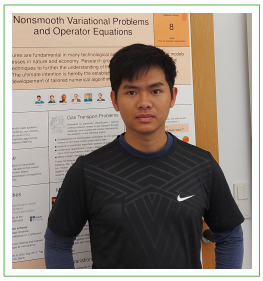

"The essence of mathematics lies in its freedom." - Georg Cantor
"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Education

2020–now **PhD student enrolled in Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis and Stochastics* (WIAS), Berlin, Germany.
In the Research Group Nonsmooth Variational Problems & Operator Equations, Head: Prof. Michael Hintermüller, within the European Innovative Training Network Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC), in the Project 11 Optimal Shape Design of Air Ducts in Combustion Engines.

2018–2019 **Master de Sciences, Technologie, Santé, Mathematiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics and Applications (skipped Master 1).

2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis and Numerical Analysis.

2011-2014 **Elementary Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

Title *Staggered and well-balanced discretization of shallow-water equations* [PDF]

Supervisor Prof. Nicolas Seguin

Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Attended Master Courses

2018-2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications, 2018-2019:**

Sep 2018 Prof. Nicoletta Tchou, Sobolev spaces & elliptic equation.

Sep 2018 Prof. Florian Méhats, Hyperbolic equations.

Nov 2018 Prof. Nicolas Seguin & Eric Darrigrand, Finite Element Method (FEM).

Nov 2018 Prof. Nicolas Seguin, Numerics of transport.

Nov 2018 Prof. Mohammed Lemou, Numerical methods for kinetic transport equations.

Dec 2018 Prof. Luis Miguel Rodrigues, Master 2 Seminar
▷ Hong Nguyen. "On the smallest constant for a Gagliardo-Nirenberg functional inequality".

Mar 2019 Vincent Duchêne, Modeling & mathematical analysis of shallow free-surface fluid flows.

Mar 2019 Prof. Marc Quincampoix, Optimal control & Hamilton-Jacobi equations.

Research Interests

- Mathematical Analysis: Partial Differential Equations (PDEs).
- Numerical Analysis: Finite Element Methods (FEMs), Finite Volume Methods (FVMs).
- Mathematical Optimization: Shape Optimization, Topology Optimization.

Experience

Vocational

2019 **Master Internship**, IRMAR, Rennes, France.

Miscellaneous

2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students* (CBT), Vietnam.
Trained the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

[1] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. "Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term". *Adv. Nonlinear Stud.* (2020). DOI https://doi.org/10.1515/ans-2020-2076.

Courses

Oct 12–23, 2020 Thematic Einstein Semester (TES): Energy-based mathematical methods for reactive multiphase flows.
▷ TES Student Compact Course: Variational Methods for Fluids and Solids.

Apr–Jun 2020 Dr. Stephan Schmidt. *Computational Geometry and Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

Seminars

Mar 2020–now Material Modeling Seminar, WIAS Berlin, Germany.

Mar 2020–now Numerical Mathematics Seminars, Research Group 3: Numerical Mathematics & Scientific Computing, WIAS Berlin, Germany.

Mar 2020–now Joint Research Seminar on Mathematical Optimization, Research Group 8: Non-smooth Variational Problems & Operator Equations, WIAS Berlin, Germany.

Mar 2020–now Berlin Mathematics Research Center MATH$^+$ Events.

Nov 2020–Apr 2021 ROMSOC Scientific Seminar.

Schools

Jul 20–24, 2020 162nd European Study Group with Industry, University of Leeds, UK.

Sep 7–10, 2020 Model Order Reduction Summer School 2020 (MORSS2020), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Workshops

Sep 28–29, 2020 ROMSOC Internal Workshop.

Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

Conferences

Sep 14–17, 2020 Deutsche Mathematiker-Vereinigung (German Mathematical Society) Annual Meeting 2020 (DMV2020), Technische Universität Chemnitz, Chemnitz, Germany.

Talks

Sep 7, 2020 Hong Nguyen (WIAS Berlin). "A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes" [Slide]. Model Order Reduction Summer School 2020 (MORSS2020), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Jun 26, 2019 Hong Nguyen. "Staggered and well-balanced discretization of shallow water equations" [Slide]. RWTH Aachen University, Germany.

Miscellaneous

Nov 30, 2020 Optimal shape design of air ducts. ROMSOC Blog.

Visits

2020 Humboldt-Universität zu Berlin, Berlin, Germany.

Jun 25–27, 2019 Department of Mathematics, Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen University, Aachen, Germany.

Awards & Scholarships

2020–2021 Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship, Innovative Training Network - European Industrial Doctorate Programs (ITN-EID), Horizon 2020.

2018–2019 Lebesgue Master Scholarship for Master 2 program.

2017 Scholarship of Vietnam Institute for Advanced Studies in Mathematics (VIASM).

2016 Special Prize; First Prize in Algebra, First Prize in Analysis, in Vietnamese Mathematical Olympiad for College Students (VMC 2016).

2015 Second Prize in Algebra, Second Prize in Analysis, in Vietnamese Mathematical Olympiad for College Students (VMC 2015).

Computer Skills

GitHub GitHub/NQBH.

Basic R, Fireshape, Firedrake, FEniCS, Python 3, Star-CCM+, Git, C/C++.

Intermediate UNIX/Linux [openSUSE, (K)Ubuntu], OpenFOAM.

Advanced LaTeX, MATLAB.

Languages

Vietnamese Mother tongue

English Intermediate

Referees

[6] **Prof. Michael Hintermüller** *Primary PhD Advisor*

[5] **Dr. Axel Kröner** *PhD Advisor*

[4] **Prof. Nicolas Seguin** *Master Advisor*

[3] **Dr. Nguyen Anh Dao** *Bachelor Supervisor*

[2] **Dr. Tan Trung Nguyen** *Bachelor Teacher*

[1] **Dr. Manh Khang Dao** *Bachelor Teacher*

`Updated to Jan 1, 2021.`